BCS-068	PARALLEL ALGORITHMS	L T P 3 1 0	4 Credits

Pre-requisites: None

Course Objectives:

CO1	Understand the concept of a pattern and the basic approach to the development of pattern recognition and machine intelligence algorithms
CO2	Understand the basic methods of feature extraction, feature evaluation, and data mining
CO3	Understand and apply both supervised and unsupervised clasification methods to detect and characterize patterns in real-world data
CO4	Develop prototype pattern recognition algorithms that can be used to study algorithm behavior and performance against real-world multivariate data
CO5	To introduce the fundamental algorithms for pattern recognition

MODULE I

Sequential model, need of alternative model, parallel computational models such as PRAM, LMCC, Hypercube, Cube Connected Cycle, Butterfly, Perfect Shuffle Computers, Tree model, Pyramid model, Fully Connected model, PRAM-CREW, EREW models, simulation of one model from another one. Performance Measures of Parallel Algorithms, speed-up and efficiency of PA, Cost- optimality, An example of illustrate Cost- optimal algorithms- such as summation, Min/Max on various models.

MODULE II

Parallel Sorting Networks, Parallel Merging Algorithms on CREW/EREW/MCC, Parallel Sorting Networks on CREW/EREW/MCC/, linear array Parallel Searching Algorithm, Kth element, Kth element in X+Y on PRAM, Parallel Matrix
Transportation and Multiplication Algorithm on PRAM, MCC, Vector-Matrix Multiplication, Solution of Linear Equation, Root finding.

MODULE III

Graph Algorithms - Connected Graphs, search and traversal, Combinatorial Algorithms- Permutation, Combinations, Derrangements.

Text Books:

1. M.J. Quinn, “Designing Efficient Algorithms for Parallel Computer”, McGrawHill.
2. S.G. Akl, “Design and Analysis of Parallel Algorithms”
3. S.G. Akl, ”Parallel Sorting Algorithm” by Academic Press

Course Outcomes: After the completion of the course the student will be able to:

CO1	Search, collect, classify and critically interpret relevant information to design a simple pattern recognition systems
CO2	Implement an advanced pattern recognition algorithm
CO3	Evaluate the result from a simple pattern recognition system
CO4	Describe and explain the difference between supervised and unsupervised learning
CO5	Become aware of the theoretical issues involved in pattern recognition system design such as the curse of dimensionality.